k-adic Collatz Constellations:
Prime-Preserving *k*-adic Collatz Chains on a Logarithmic Spiral

March 10, 2026 — Hiroshi Harada

1. Overview

This project visualizes prime-preserving sequences generated by *k*-adic Collatz-type maps on a logarithmic spiral, revealing hidden geometric symmetries within these prime chains. In a logarithmic polar coordinate system, each integer n is mapped as:

$$r = \ln(n)$$

$$\theta = 2\pi \ln(n)$$

In this space, the geometric shape of a sequence is determined directly by its multiplicative growth rate. If a recurrence multiplies n by approximately *a*, the angular displacement satisfies:

$$\Delta\theta \approx 2\pi \ln(a)$$

When $\ln(a)$ lies extremely close to a rational number p/q, the sequence rotates nearly p full turns every q steps, producing self-similar expanding geometric figures ("constellations"). In this study, we consider the general *k*-adic Collatz-type map:

$$F(n) = ((k + 1)n + b) / k$$

Here, b is chosen according to the residue $n \bmod k$ so that the numerator $(k + 1)n + b$ becomes divisible by k. Prime-preserving chains from *2*-adic to *7*-adic systems were explored and visualized.

2. *k*-adic Collatz Constellations

All chains listed below consist entirely of prime numbers, verified using SymPy's deterministic primality test.

2.1 *2*-adic Collatz Constellations (Pentagram)

Growth factor: $a = 3/2$

$\ln(a) = \ln(3/2) \approx 0.405$ → Close to $2/5$ → about two turns every five steps → pentagram-like pattern.

Prime chain (L9):

35014031359 → 52521047039 → 78781570559 → 118172355839 → 177258533759 → 265887800639 → 398831700959 → 598247551439 → 897371327159

2.2 *3*-adic Collatz Constellations (Heptagram)

Growth factor: $a = 4/3$

$\ln(a) = \ln(4/3) \approx 0.288$ → Close to $2/7$ → two turns every seven steps → heptagram-like pattern.

Prime chain (L8):

46000363 → 61333817 → 81778423 → 109037897 → 145383863 → 193845151 → 258460201 → 344613601

2.3 *4*-adic Collatz Constellations (Enneagram)

Growth factor: $a = 5/4$

$\ln(a) = \ln(5/4) \approx 0.223$ → Close to $2/9$ → two turns every nine steps → enneagram-like pattern.

Prime chain (L7):

559188056938807297 → 698985071173509121 → 873731338966886401 → 1092164173708608001 → 1365205217135760001 → 1706506521419700001 → 2133133151774625001

2.4 *5*-adic Collatz Constellations (Hendecagram)

Growth factor: $a = 6/5$

$ln(a) = ln(6/5) \approx 0.182$ → Close to $2/11$ → two turns every eleven steps → hendecagram-like pattern.

Prime chain (L7):

76933159 → 92319791 → 110783749 → 132940499 → 159528599 → 191434319 → 229721183

2.5 *6*-adic Collatz Constellations (Tridecagram)

Growth factor: $a = 7/6$

$ln(a) = ln(7/6) \approx 0.154$ → Close to $2/13$ → two turns every thirteen steps → tridecagram-like pattern.

Prime chain (L8):

1099687 → 1282969 → 1496797 → 1746263 → 2037307 → 2376859 → 2773003 → 3235171

2.6 *7*-adic Collatz Constellations (Pentadecagram)

Growth factor: $a = 8/7$

$ln(a) = ln(8/7) \approx 0.134$ → Close to $2/15$ → two turns every fifteen steps → pentadecagram-like pattern.

Prime chain (L9):

68542687 → 78334499 → 89525141 → 102314447 → 116930797 → 133635197 → 152725939 → 174543931 → 199478779

3. Results

Across the range from *2*-adic to *7*-adic systems, all discovered sequences were confirmed to be prime-preserving Collatz chains. The *4*-adic chains include 18–19 digit primes, all verified using SymPy's deterministic primality test. When plotted on a logarithmic spiral, each *k*-adic system exhibits a distinct rotational period, producing a diverse family of "prime constellations" that highlight the geometric nature of *k*-adic Collatz dynamics.

License

© 2026 Hiroshi Harada — MIT License.
This work and the accompanying code are licensed under the MIT License.

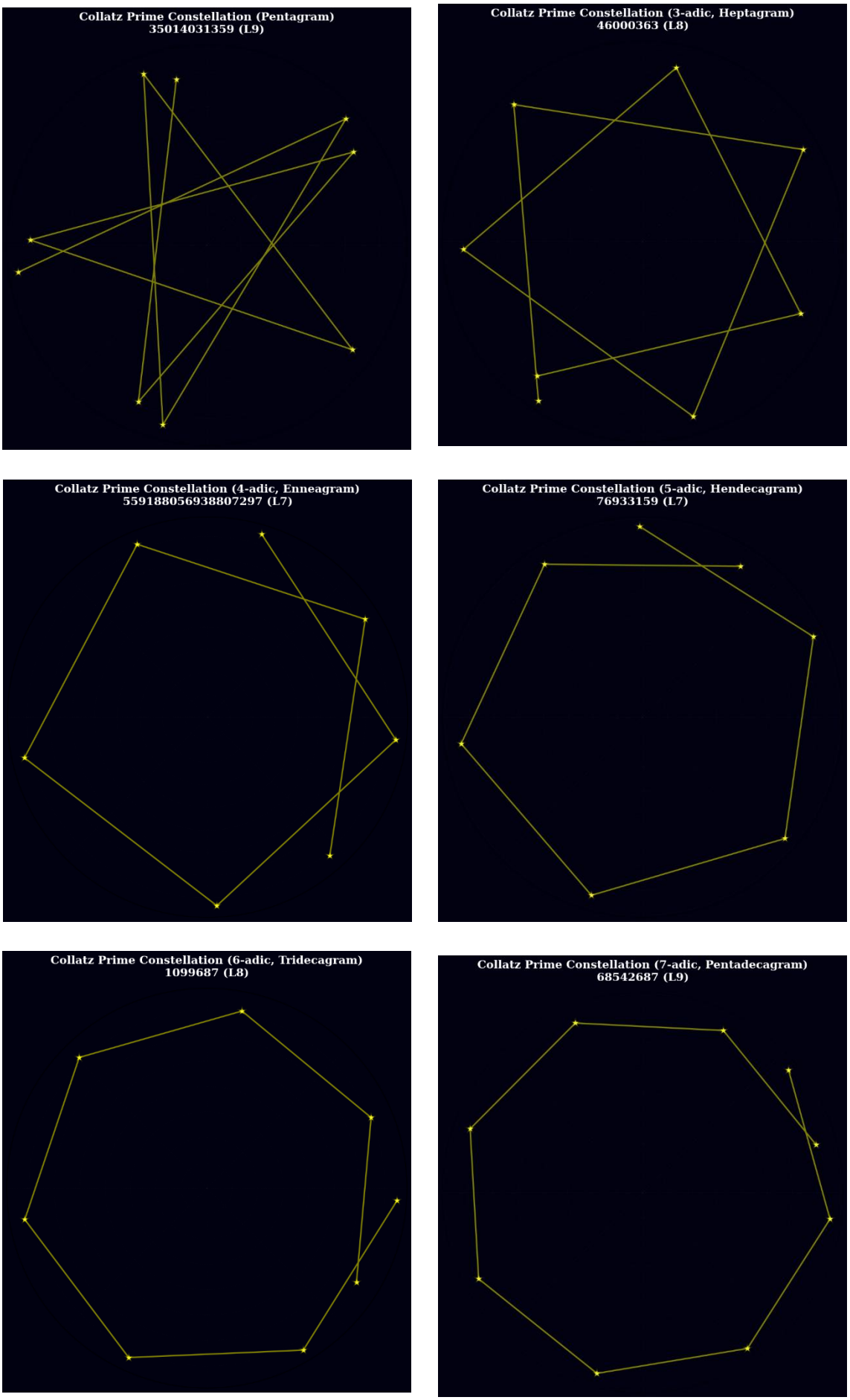

Figure 1. *k*-adic Collatz Prime Chain Constellations.